HDOC Day 43 - C Programming

Question

Write a menu-driven C program using separate functions to (1) find the longest word in a sentence and (2) check whether one string is a rotation of another.

C Program

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

void longestWord() {
    char str[200], word[200] = "";
    int i, start = 0, len = 0, max = 0;

    printf("Enter a sentence: ");
    fgets(str, sizeof(str), stdin);

    for (i = 0; ; i++) {
        if (isalnum((unsigned char)str[i])) {
            if (len == 0) start = i;
            len++;
        } else {
            if (len > max) {
                max = len;
                strncpy(word, str + start, len);
                word[len] = '\0';
            }
            len = 0;
            if (str[i] == '\0') break;
        }
    }
    if (max == 0) printf("No word found.\n");
    else printf("Longest word: %s (Length: %d)\n", word, max);
}

void checkRotation() {
    char a[100], b[100], joined[200];
    printf("Enter first string: ");
    fgets(a, sizeof(a), stdin);
    printf("Enter second string: ");
    fgets(b, sizeof(b), stdin);
    a[strcspn(a, "\n")] = '\0';
    b[strcspn(b, "\n")] = '\0';

    if (strlen(a) != strlen(b)) {
        printf("Not a rotation.\n");
        return;
    }
    strcpy(joined, a);
    strcat(joined, a);
    if (strstr(joined, b) != NULL)
        printf("The strings are rotations.\n");
    else
        printf("Not a rotation.\n");
}

int main() {
    int choice;
    char line[30];
    do {
        printf("\n1. Find longest word\n");
```

```
        printf("2. Check string rotation\n");
        printf("3. Exit\nEnter choice: ");
        if (!fgets(line, sizeof(line), stdin)) break;
        if (sscanf(line, "%d", &choice) != 1) choice = 0;
        switch (choice) {
            case 1: longestWord(); break;
            case 2: checkRotation(); break;
            case 3: printf("Program ended.\n"); break;
            default: printf("Invalid choice.\n");
        }
    } while (choice != 3);
    return 0;
}
```

Sample Output

```
1. Find longest word
2. Check string rotation
3. Exit
Enter choice: 1
Enter a sentence: I love programming
Longest word: programming (Length: 11)

Enter choice: 2
Enter first string: ABCD
Enter second string: CDAB
The strings are rotations.

Enter choice: 3
Program ended.
```

Explanation

The first function checks the words in a sentence and prints the longest one. The second function joins the first string with itself and checks whether the second string appears in it. The menu repeats until the user chooses Exit.